

Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is limited by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts across subjects and sessions. We propose a domain-adaptive selective state-space framework for this setting. A multi-kernel temporal convolutional front-end with depthwise spatial filtering and channel-group attention extracts local spectral-spatial features. Five selective state-space blocks with alternating forward and backward scans then model long-range temporal context at a cost linear in sequence length, and a convolutional shortcut with a grouped dilated temporal convolutional network fuses local and global information. To reduce cross-subject shift, the framework combines per-subject Riemannian alignment, an automatic domain alignment (AutoDIAL) layer that learns how strongly target features are normalized with target rather than source statistics, gradient-reversal adversarial learning, and multi-kernel maximum mean discrepancy at the temporal and trial levels. At inference time, an information-maximization stage adapts only normalization parameters using unlabeled evaluation-session trials and re-estimates the AutoDIAL target statistics; their labels remain hidden until scoring. Across nine LOSO target subjects, Ours achieved $75.81 \pm 10.63\%$ session-2 accuracy on BCI IV-2a and $83.41 \pm 10.35\%$ sessions 4–5 accuracy on BCI IV-2b, with Cohen’s κ scores of 67.70 ± 14.20 and 66.80 ± 20.70 , respectively, on the $\times 100$ scale. Across the four Zhou2016 targets, it reached $74.00 \pm 6.02\%$ session-3 accuracy ($\kappa = 61.00 \pm 9.03$). Code will be released at <https://github.com/USERNAME/REPOSITORY>.

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1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG), with applications in rehabilitation, assistive communication, and human–machine control. Their principal signatures, event-related desynchronization and synchronization, are weak, non-stationary, and distributed across time and electrodes [1]. They also vary substantially among subjects and recording sessions. Consequently, a model trained on existing users often degrades on an unseen user, whereas acquiring labeled calibration trials for every user is time-consuming. Cross-subject decoding must therefore learn discriminative spatial–temporal structure from limited EEG while controlling inter-subject distribution shift without target labels.

Classical MI pipelines rely on common spatial pattern (CSP), which maximizes class-specific variance, and filter-bank CSP (FBCSP), which extends this idea across frequency bands [2, 3]. Although effective and interpretable, these methods separate filtering, feature extraction, and classification into manually designed stages; conventional CSP is also naturally binary and sensitive to the available trials and frequency range. Their subject-specific covariance structure can therefore transfer poorly when the user changes.

End-to-end convolutional neural networks reduce this dependence on handcrafted processing. Deep and shallow ConvNets learn temporal and electrode filters directly from EEG, while EEGNet uses depthwise and separable convolutions to obtain compact frequency-specific spatial features [4, 5]. Multi-scale temporal kernels are especially relevant because MI information is not confined to one frequency or duration. However, convolutions connect distant time points only through additional depth, larger kernels, or dilation, which may be insufficient to characterize the full temporal evolution of a trial.

Hybrid decoders therefore combine EEG-specific convolution with global sequence modeling. EEG Conformer applies self-attention after convolutional tokenization, ATCNet combines attention with a dilated temporal convolutional network (TCN), and TCFormer integrates a multi-kernel front-end,

grouped-query Transformer layers, and a grouped TCN head [6, 7, 8]. These models demonstrate the value of combining local spectral-spatial priors with longer temporal context. Nevertheless, self-attention explicitly constructs pairwise token interactions and has quadratic sequence complexity [9]. This cost becomes undesirable for long, high-resolution EEG, while highly flexible attention operators may also be data-hungry in the small-sample regimes common in BCI studies.

Architecture alone cannot remove subject shift. Target-free domain generalization learns invariant representations from source subjects [10, 11, 12], whereas unsupervised domain adaptation (UDA) may additionally exploit unlabeled target EEG. The distinction is important: observing target signals without labels is UDA rather than zero-shot generalization. Existing UDA methods reduce covariance mismatch through Riemannian or Euclidean alignment [13, 14], minimize feature discrepancy with MK-MMD [15], or promote domain confusion through gradient reversal [16, 17]. HADANet combines adversarial and MK-MMD alignment for MI-EEG [18], while prototype-based methods additionally exploit local class structure [19]. Domain shift can also be reduced inside normalization layers: AutoDIAL standardizes target features with a learned mixture of source and target statistics, so the network itself determines how much domain-specific normalization is beneficial [20]. These approaches remain vulnerable to negative transfer: marginal alignment may mix classes, and aligning only pooled trial features can overlook temporal mismatch.

Residual shift may also be addressed at inference. Information-maximization methods encourage confident but globally diverse predictions without labels [21], and TENT restricts adaptation to normalization parameters through entropy minimization [22]. Such constraints are attractive for EEG because target datasets are small and unrestricted fine-tuning can overfit or collapse to one predicted class. However, evaluation-session access must be disclosed explicitly, and labels must remain unavailable during adaptation and checkpoint selection.

Selective state-space models offer a linear-complexity alternative to attention. S4 introduced structured long-range sequence modeling [23], and Mamba made transition quantities input-dependent so that relevant information can be retained selectively [24]. Recent MI decoders have combined convolution and Mamba [25, 26], but primarily study architecture rather than leakage-controlled integration with target adaptation. Moreover, a generic state-space stack should complement, rather than replace, explicit tempo-

ral and electrode filtering. A purely forward scan also imposes unnecessary directionality when complete offline MI trials are available.

We therefore present an improved cross-subject MI-EEG decoding framework that jointly addresses temporal representation and subject shift. The framework learns complementary local and long-range EEG dynamics, reduces distribution mismatch during training, and performs constrained label-free adaptation at inference. Target labels remain hidden throughout preprocessing, training, checkpoint selection, and adaptation.

The contributions of this study are:

- an improved end-to-end framework for cross-subject MI-EEG decoding that unifies multi-scale local representation, selective long-range temporal modeling, and local-global feature fusion;
- a hierarchical adaptation strategy with a chronological, leakage-controlled LOSO protocol that separates source learning, unlabeled target adaptation, and held-out evaluation; and
- a comprehensive evaluation on three public MI-EEG datasets using subject-wise accuracy, Cohen’s κ , efficiency analysis, and component ablations.

2. Datasets

Three public MI-EEG datasets are used to examine whether the same decoder transfers across different numbers of electrodes, classes, subjects, and recording sessions. Dataset access and event extraction are implemented through the MOABB/Braindecode interface; this choice provides a common representation of subjects, sessions, runs, and event windows and supports reproducible benchmarking [27]. Table 1 summarizes the properties relevant to the present work.

2.1. BCI Competition IV-2a

BCI Competition IV dataset 2a contains recordings from nine participants. Each participant completed two sessions on different days, and each session contains 288 trials distributed equally over imagined left-hand, right-hand, both-feet, and tongue movements. Signals were acquired from 22 EEG electrodes and three EOG electrodes at 250 Hz; only the EEG electrodes are supplied to the classifier. The original recordings were hardware band-pass

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Sex (M/F)	Channels	Sessions	Classes
BCI IV-2a	9	n.r.	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	n.r. ^a	3	5	left hand, right hand
Zhou2016	4	1/3	14	3	left hand, right hand, feet

n.r.: not reported in the dataset description. ^aThe nine participants were drawn from a study of ten (6/4, 24.7 ± 3.3 years) [28]; the split of the nine released participants is not reported.

filtered between 0.5 and 100 Hz and notch filtered at 50 Hz [29]. The implementation uses the event-defined MI interval exposed by MOABB, rescales volts to microvolts, and does not apply an additional frequency filter. The first recording session is denoted session 1 and the second is denoted session 2 throughout this paper. The dataset description does not report participant sex or age.

2.2. BCI Competition IV-2b

BCI Competition IV dataset 2b contains two-class left- versus right-hand MI recordings from nine right-handed participants with normal or corrected-to-normal vision, drawn from a study of ten participants (six male, four female; age 24.7 ± 3.3 years) [28]; the sex of the nine released participants is not reported individually. Three bipolar EEG derivations centered on C3, Cz, and C4 were sampled at 250 Hz, together with three EOG channels that are excluded from decoding. Each participant completed five sessions: the first three form the calibration portion and the final two form the original evaluation portion [29]. The present loader follows this natural chronological separation. The event-defined MI window is used after removing the final 0.5 s through the configured stop offset. This dataset tests whether the method remains useful when spatial coverage is reduced from 22 electrodes to only three.

2.3. Zhou2016

The Zhou2016 dataset contains four healthy participants (one male and three female, aged 22–28 years), each recorded in three sessions separated by days to months; sex is reported only for the group, not per participant [30]. A session consists of two runs of 75 trials, with 25 trials per class, giving 450

trials per participant. The three MI classes are left hand, right hand, and feet. EEG was recorded from 14 electrodes at 250 Hz. Each trial contains a 5-s visual arrow period during which the indicated imagery is performed, followed by a 4-s rest period [30]. The public data are loaded through the MOABB representation [27]; the dataset-specific frequency and time-window processing is described in Section 4.2.

3. Proposed Method

Figure 1 summarizes one LOSO fold. A shared encoder processes labeled source trials and unlabeled target trials during Stage 1 UDA; an AutoDIAL layer then normalizes the pooled trial features before the residual aligner, classifier, and alignment losses. After source-validation checkpoint selection, Stage 2 adapts only normalization parameters with unlabeled evaluation-session EEG and re-estimates the AutoDIAL target statistics; labels are revealed only for Stage 3 scoring.

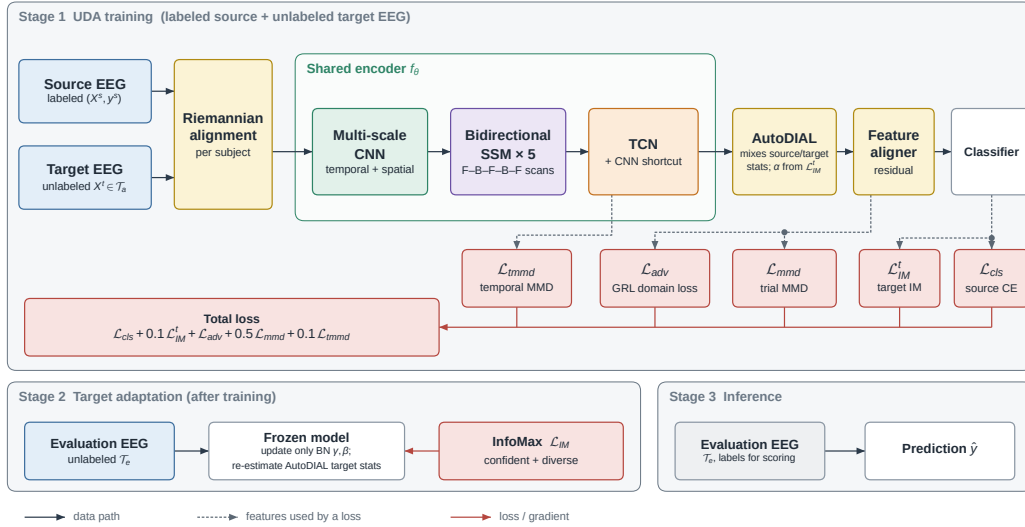


Figure 1: Framework for one LOSO fold: joint source classification and target adaptation with AutoDIAL feature normalization (Stage 1), label-free IM-TTA followed by AutoDIAL target-statistic re-estimation on the evaluation sessions $\mathcal{T}_e$ (Stage 2), and scoring with previously withheld labels (Stage 3).

3.1. Problem formulation

Let $\mathcal{S} = \{(X_i^s, y_i^s)\}_{i=1}^{N_s}$ be labeled trials from the source subjects, $\mathcal{T}_a = \{X_j^a\}_{j=1}^{N_a}$ the earlier unlabeled target sessions used for UDA, and $\mathcal{T}_e = \{X_j^e\}_{j=1}^{N_e}$ the later target sessions used for evaluation, where $X \in \mathbb{R}^{C \times T}$. Training uses $\mathcal{S}$ and $\mathcal{T}_a$; IM-TTA may access only the signal values of $\mathcal{T}_e$, never their labels.

3.2. Leakage-safe signal alignment

A separate Riemannian whitening operator is optionally fitted for each subject. OAS trial covariances are summarized by their affine-invariant mean $\bar{R}$, and trials are transformed as

$$\tilde{X} = \bar{R}^{-1/2} X. \quad (1)$$

A source transform is fitted on that subject’s training sessions and applied to its validation sessions; the target transform is fitted on $\mathcal{T}_a$ and frozen before application to $\mathcal{T}_e$ [13, 14]. Evaluation-session data therefore update neither the Riemannian transforms nor the input z-score statistics; the only statistics estimated on $\mathcal{T}_e$ are the AutoDIAL target moments in Stage 2 (Section 3.8).

3.3. Multi-scale convolutional encoder

As detailed in Fig. 2, three temporal convolutions with kernels of 20, 32, and 64 samples produce 32 maps each [8]. Their concatenated output is processed by depthwise convolution across all electrodes with depth multiplier two, average pooling, and dropout. A grouped pointwise projection reduces the 192 spatial maps to 48 channels (16 per temporal scale), followed by grouped temporal convolution and residual channel-group attention. The latter assigns one learned coefficient to each scale. A second pooling layer yields $H_c \in \mathbb{R}^{B \times 48 \times L}$.

3.4. Alternating selective state-space encoder

After pointwise channel mixing, five selective state-space blocks scan in the order forward–backward–forward–backward–forward. A backward block reverses the temporal axis before and after its recurrence, providing bidirectional offline context without duplicating every block (Fig. 3). Within each

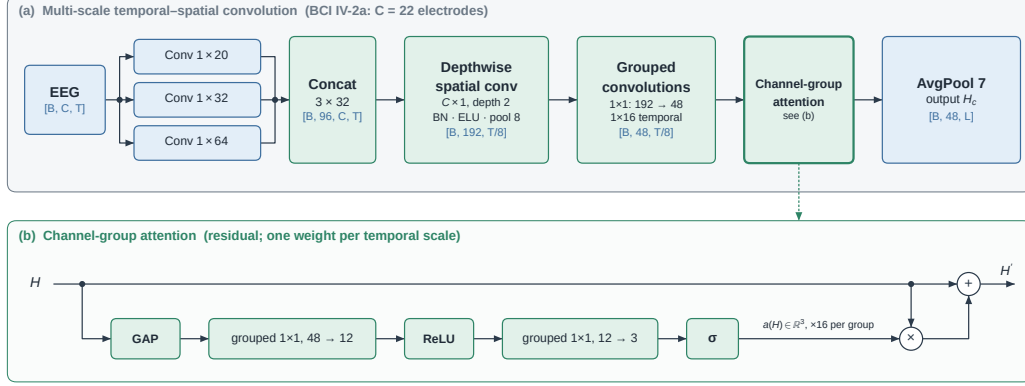


Figure 2: Multi-scale temporal-spatial front-end. (a) Three temporal branches, depth-wise electrode filtering, grouped refinement, and channel-group attention produce H_c . (b) Channel-group attention: grouped 1×1 squeeze-excitation yields one sigmoid weight per temporal scale, which rescales that scale’s 16 channels and is added residually.

block, layer normalization, gated projections, and a causal depthwise convolution generate u_t . A diagonal selective recurrence then computes [23, 24]

$$A = -\exp(A_{\log}), \quad (2)$$

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \quad (3)$$

$$z_t = C_t^\top h_t + d_{\text{skip}} \odot u_t. \quad (4)$$

Here Δ_t , B_t , and C_t depend on the input. The implementation uses state dimension eight, dropout 0.4, and depth-dependent stochastic depth. It is a lightweight PyTorch selective SSM rather than the official hardware-aware Mamba kernel, but retains linear scaling in sequence length.

3.5. Local-global fusion and classification

The state-space output is reduced from 48 to 16 channels and concatenated with the 48-channel shortcut H_c . Two residual grouped TCN blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The mean of its last state and temporal average forms the trial representation. A norm-constrained grouped 1×1 classifier produces four group-wise logits, which are combined by learned softmax weights.

3.6. Automatic domain alignment layer

Let $q \in \mathbb{R}^{64}$ denote a pooled trial representation, formed by the 16-channel groups of the three convolutional scales and the state-space branch. Following

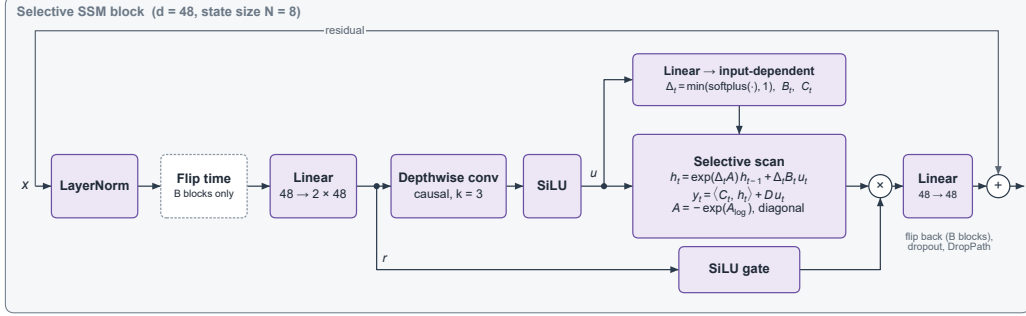


Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

AutoDIAL [20], an alignment layer standardizes source features with source statistics (μ_s, σ_s^2) , whereas target features are standardized with the moments of the source–target mixture weighted by α :

$$\mu_\alpha = \alpha\mu_t + (1 - \alpha)\mu_s, \quad (5)$$

$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1 - \alpha)\sigma_s^2 + \alpha(1 - \alpha)(\mu_t - \mu_s)^2, \quad (6)$$

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \quad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta. \quad (7)$$

The affine parameters γ and β are shared by both domains. A separate $\alpha = \text{sigmoid}(a) \in (0, 1)$, initialized to 0.5, is learned for each of the four feature groups: $\alpha \rightarrow 0$ normalizes the target with source statistics, whereas $\alpha \rightarrow 1$ recovers fully domain-specific normalization. During training, mini-batch moments of each domain are used and running estimates are updated with momentum 0.1; at inference, the running source and target moments are mixed with the learned α . Source validation trials are always normalized with source statistics.

3.7. Hierarchical unsupervised domain adaptation

A residual $64 \rightarrow 128 \rightarrow 64$ feature aligner follows the AutoDIAL layer and precedes classification and domain alignment. Source cross-entropy $\mathcal{L}_{\text{cls}}$ maintains discrimination. A gradient-reversal discriminator provides domain loss $\mathcal{L}_{\text{adv}}$ [16], while five-kernel MK-MMD aligns normalized source and target representations [15]. It is applied both to pooled trial features ($\mathcal{L}_{\text{mmd}}$) and to time-averaged TCN sequences ($\mathcal{L}_{\text{tmmd}}$). In addition, the information-maximization objective $\mathcal{L}_{\text{IM}}$ of Eq. (9) is evaluated on the predictions for the

unlabeled target mini-batch ($\mathcal{L}_{\text{IM}}^t$), giving

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}_{\text{IM}}^t + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \quad (8)$$

The alignment losses $\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{mmd}}$, and $\mathcal{L}_{\text{tmmd}}$ see the AutoDIAL output with α held fixed, because they could otherwise be reduced trivially by driving α toward one. Since source features do not depend on α , the mixing weights are learned only through $\mathcal{L}_{\text{IM}}^t$, that is, toward the normalization that yields confident yet class-diverse target predictions. Only $\mathcal{L}_{\text{cls}}$ uses labels, all of which come from source subjects; target trials in this stage come exclusively from $\mathcal{T}_a$. The discriminator coefficient increases smoothly from zero to one with training progress.

3.8. Constrained information-maximization adaptation

After checkpoint selection, all parameters except BatchNorm affine weights and biases are frozen. Five Adam passes over unlabeled $\mathcal{T}_e$ at learning rate 10^{-4} minimize

$$\mathcal{L}_{\text{IM}} = \frac{1}{N_e} \sum_j H(p_j) - H\left(\frac{1}{N_e} \sum_j p_j\right), \quad (9)$$

where the first term encourages confident predictions and the second preserves class diversity [21, 22]. Because IM-TTA changes the backbone features, the AutoDIAL target moments (μ_t, σ_t^2) are then re-estimated in one pass over the same unlabeled $\mathcal{T}_e$ trials, while the learned α and the source moments remain fixed. No evaluation label or metric is used for adaptation, re-estimation, or checkpoint selection.

4. Experimental Protocol

4.1. Chronological LOSO protocol with test-time adaptation

One subject is held out in each LOSO fold. All remaining subjects are sources; the held-out participant is the target. Table 2 specifies the chronological session split. Earlier target sessions are used without labels during Stage 1 UDA. The later evaluation sessions remain unavailable during model training, preprocessing-statistic fitting, and checkpoint selection; at inference time, their EEG signals are passed without labels to IM-TTA and AutoDIAL target-statistic re-estimation, after which the adapted model is scored using the withheld labels. This is the primary protocol for headline comparisons.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session EEG is available without labels only to IM-TTA and AutoDIAL target-statistic re-estimation at inference time; its labels are used afterward solely for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	sessions 1–2	session 3

For BCI IV-2a, this split yields nine folds; for BCI IV-2b, nine folds; and for Zhou2016, four folds. The target identifier changes between folds, but the split rule and hyperparameters remain fixed. For Zhou2016, source sessions 1–2 are pooled for training and source session 3 is used only for validation; target sessions 1–2 are pooled as unlabeled Stage 1 adaptation data and target session 3 is reserved for IM-TTA and primary scoring. The checkpoint with the highest source-validation accuracy is selected before IM-TTA. Neither evaluation-session labels nor evaluation metrics participate in adaptation or checkpoint selection. Stage 1 AutoDIAL, target information-maximization, and adversarial/MK-MMD training use the earlier unlabeled target-adaptation loader, whereas Stage 2 IM-TTA and AutoDIAL re-estimation use the later unlabeled evaluation loader immediately before predictions are scored. In addition to the primary Zhou2016 session-3 metric, an all-session target score is retained only as an explicitly auxiliary diagnostic.

4.2. Preprocessing and augmentation

Only EEG channels are retained. Signals are resampled to 250 Hz and converted from volts to microvolts. The BCI Competition datasets receive no additional digital band-pass filtering: BCI IV-2a uses the complete MOABB event window, while BCI IV-2b removes 0.5 s from its end. For Zhou2016, all 14 EEG channels are band-pass filtered at 8–30 Hz and the interval from 0.5 to 5.0 s after cue onset is retained (4.5 s, or 1125 samples). This setting suppresses the initial cue transient and focuses both feature extraction and covariance estimation on the motor-imagery μ/β range.

Riemannian whitening, when active, follows the session-isolated fitting rule in Section 3.2. Z-score statistics are fitted using the source-training partition and then frozen. Inter-trial segment reconstruction augmentation is applied only to labeled source mini-batches: segments are recombined among trials of the same source class, so it neither reads target labels nor creates mixed source–target examples. Zhou2016 is run without this augmentation to preserve its intended dataset-specific setting.

4.3. Training configuration

All experiments are executed in Kaggle notebooks using one NVIDIA Tesla T4 GPU with 16 GB memory. BCI IV-2a and BCI IV-2b use a mini-batch size of 48 and a maximum of 125 epochs. Zhou2016 uses a mini-batch size of 32 and 100 epochs. The BCI Competition runs use Adam with an initial learning rate of 9×10^{-4} , $\beta_1 = 0.5$, and weight decay 10^{-3} ; the Zhou2016 setting uses an initial learning rate of 10^{-3} and weight decay 10^{-4} . A linear warm-up over the first three LOSO epochs is followed by cosine decay. The random seed (0), data split, and initialization procedure are fixed and shared by all methods and ablation variants, so that every reported comparison differs only in the evaluated component.

4.4. Evaluation measures and reporting

Accuracy and Cohen’s κ ($\times 100$) are reported per target subject and as mean $\pm$ standard deviation across LOSO folds. Paired accuracy differences use a two-sided Wilcoxon signed-rank test with exact sign-permutation p -values. Efficiency is summarized by trainable parameters, mean Stage 1 training time per fold on a Tesla T4, and batch-one CPU response time averaged over 500 passes after 100 warm-up passes; IM-TTA, preprocessing, and data transfer are excluded from these timings.

4.5. Baseline comparison policy

All methods follow the same LOSO folds, chronological split, and target-data access in Table 2; results obtained under incompatible protocols are not copied. EEG Conformer [6], ATCNet [7], TCFormer [8], DAN [15], and CDAN [17] are the selected baselines. The proposed method is reported in three variants:

- **Ours-S**: source-only training with $\mathcal{L}_{\text{cls}}$;

- **Ours-U**: Stage 1 UDA, including AutoDIAL, without IM-TTA; the AutoDIAL target statistics are still re-estimated on unlabeled $\mathcal{T}_e$; and
- **Ours**: Ours-U followed by label-free IM-TTA on the evaluation-session EEG.

Thus, Ours-U–Ours-S isolates Stage 1 adaptation, while Ours–Ours-U isolates IM-TTA.

5. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

5.1. BCI Competition IV-2a

Table 3 reports the four-class chronological-session LOSO results for the nine BCI IV-2a target subjects. Ours applies label-free IM-TTA to each target subject’s session 2 EEG immediately before evaluation.

Ours achieved $75.81 \pm 10.63\%$ accuracy and 67.70 ± 14.20 Cohen’s κ , outperforming Ours-S ($64.35 \pm 12.77\%$) and Ours-U ($72.15 \pm 13.58\%$). Both paired accuracy differences were significant ($p = 0.0039$ and $p = 0.0078$, respectively); IM-TTA improved eight of the nine target folds. As an auxiliary diagnostic only, Ours-U obtained $73.32 \pm 11.94\%$ accuracy and 64.40 ± 15.90 Cohen’s κ when target sessions 1 and 2 were pooled.

5.2. BCI Competition IV-2b

Table 4 reports the completed binary left- versus right-hand results for the nine BCI IV-2b target subjects, whose recordings contain only three bipolar EEG derivations. Results use sessions 4–5 for the primary chronological evaluation; the standard deviation measures between-subject dispersion in this single LOSO run.

Ours achieved $83.41 \pm 10.35\%$ accuracy and 66.80 ± 20.70 Cohen’s κ , compared with $80.54 \pm 10.07\%$ for Ours-S and $83.97 \pm 10.71\%$ for Ours-U. Ours-U was 0.56 percentage points higher than Ours on average, but the paired difference was not significant ($p = 0.4063$); Ours likewise did not differ significantly from Ours-S ($p = 0.2500$). IM-TTA improved three of the nine target folds.

Table 3: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2a (four classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	—	—	—	68.06	—	—	79.17	78.82
	κ	—	—	—	57.41	—	—	72.22	71.76
S2	Acc.	—	—	—	51.04	—	—	47.57	57.64
	κ	—	—	—	34.72	—	—	30.09	43.52
S3	Acc.	—	—	—	86.46	—	—	92.71	93.75
	κ	—	—	—	81.94	—	—	90.28	91.67
S4	Acc.	—	—	—	61.11	—	—	73.61	74.31
	κ	—	—	—	48.15	—	—	64.81	65.74
S5	Acc.	—	—	—	64.24	—	—	57.64	65.62
	κ	—	—	—	52.31	—	—	43.52	54.17
S6	Acc.	—	—	—	39.58	—	—	59.72	66.67
	κ	—	—	—	19.44	—	—	46.30	55.56
S7	Acc.	—	—	—	72.92	—	—	82.29	85.42
	κ	—	—	—	63.89	—	—	76.39	80.56
S8	Acc.	—	—	—	73.61	—	—	82.64	84.38
	κ	—	—	—	64.81	—	—	76.85	79.17
S9	Acc.	—	—	—	62.15	—	—	73.96	75.69
	κ	—	—	—	49.54	—	—	65.28	67.59
Mean	Acc.	—	—	—	64.35	—	—	72.15	75.81
	κ	—	—	—	52.50	—	—	62.90	67.70
SD	Acc.	—	—	—	12.77	—	—	13.58	10.63
	κ	—	—	—	17.00	—	—	18.10	14.20
p		—	—	—	0.0039	—	—	0.0078	—

Table 4: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2b (two classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	—	—	—	77.50	—	—	78.44	77.50
	κ	—	—	—	55.00	—	—	56.88	55.00
S2	Acc.	—	—	—	68.57	—	—	70.71	67.86
	κ	—	—	—	37.14	—	—	41.43	35.71
S3	Acc.	—	—	—	63.13	—	—	63.44	65.94
	κ	—	—	—	26.25	—	—	26.88	31.88
S4	Acc.	—	—	—	89.38	—	—	90.31	88.75
	κ	—	—	—	78.75	—	—	80.62	77.50
S5	Acc.	—	—	—	96.56	—	—	97.81	96.88
	κ	—	—	—	93.12	—	—	95.63	93.75
S6	Acc.	—	—	—	73.44	—	—	83.75	85.94
	κ	—	—	—	46.88	—	—	67.50	71.88
S7	Acc.	—	—	—	85.00	—	—	89.69	90.62
	κ	—	—	—	70.00	—	—	79.37	81.25
S8	Acc.	—	—	—	86.25	—	—	95.63	94.06
	κ	—	—	—	72.50	—	—	91.25	88.13
S9	Acc.	—	—	—	85.00	—	—	85.94	83.13
	κ	—	—	—	70.00	—	—	71.88	66.25
Mean	Acc.	—	—	—	80.54	—	—	83.97	83.41
	κ	—	—	—	61.10	—	—	67.90	66.80
SD	Acc.	—	—	—	10.07	—	—	10.71	10.35
	κ	—	—	—	20.10	—	—	21.40	20.70
p		—	—	—	0.2500	—	—	0.4063	—

5.3. Zhou2016

Table 5 uses source sessions 1–2 for training, source session 3 for validation, target sessions 1–2 for UDA, and target session 3 for primary scoring. With four folds, the smallest attainable two-sided Wilcoxon p -value is 0.125.

Table 5: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on target session 3 of Zhou2016 (three classes) under the clean chronological protocol. Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation. Bold: best value in each row. p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	–	–	68.00	72.00	–	–	79.33	82.00
	κ	–	–	52.00	58.00	–	–	69.00	73.00
S2	Acc.	–	–	78.67	78.67	–	–	84.67	70.00
	κ	–	–	68.00	68.00	–	–	77.00	55.00
S3	Acc.	–	–	66.67	71.33	–	–	87.33	66.67
	κ	–	–	50.00	57.00	–	–	81.00	50.00
S4	Acc.	–	–	80.67	78.67	–	–	78.00	77.33
	κ	–	–	71.00	68.00	–	–	67.00	66.00
Mean	Acc.	–	–	73.50	75.17	–	–	82.33	74.00
	κ	–	–	60.25	62.75	–	–	73.50	61.00
SD	Acc.	–	–	6.22	3.51	–	–	3.82	6.02
	κ	–	–	9.34	5.26	–	–	5.70	9.03
p		–	–	1.0000	0.8750	–	–	0.3750	–

Ours reached $74.00 \pm 6.02\%$ accuracy and 61.00 ± 9.03 Cohen’s κ . The matched TCFormer baseline obtained $73.50 \pm 6.22\%$ and 60.25 ± 9.34 , respectively ($p = 1.0000$). Ours-U, which omits IM-TTA while retaining Stage 1 adaptation and AutoDIAL target-statistic re-estimation, obtained the highest mean accuracy, $82.33 \pm 3.82\%$ ($\kappa = 73.50 \pm 5.70$); its four-fold difference from Ours was not significant ($p = 0.3750$). Ours-S achieved $75.17 \pm 3.51\%$ ($\kappa = 62.75 \pm 5.26$; $p = 0.8750$). As an auxiliary diagnostic only, TCFormer obtained $75.91 \pm 4.65\%$ accuracy and 63.90 ± 7.00 Cohen’s κ when all three target sessions were pooled.

5.4. Computational efficiency

Table 6 reports trainable parameters, Stage 1 training time, and CPU response time. Ours-S is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

Because the token sequence after temporal pooling is short (20 tokens on Zhou2016), a step-by-step selective scan is dominated by per-operation dispatch overhead rather than arithmetic. The Zhou2016 Ours-S, Ours-U, and w/o-RA runs therefore use a loop-hoisted scan: $\exp(\Delta_t A)$, $\Delta_t B_t \odot u_t$, and the C_t readout are computed for all time steps outside the recurrence, leaving one multiply-add per step, and the causal depthwise convolution is evaluated as shifted multiply-adds. This reorganization changes neither the model nor its parameters (output logits agree with the per-step scan to within 10^{-8}). Ours-S required 6.29 min per fold and 13.36 ms per CPU trial, compared with 11.42 min and 14.65 ms for Ours-U and 11.66 min and 18.88 ms for w/o RA. The remaining Zhou2016 Ours rows have not yet been re-measured with this implementation.

Table 6: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. [†]Measured with the loop-hoisted selective scan (Section 5.4); the other Ours rows use the per-step scan, which computes the same function.

Dataset	Method	Params (K)	Train (min)	CPU (ms)
BCI IV-2a	Ours-S	98.4	47.01	25.08
	Ours-U	132.0	58.45	18.44
	Ours w/o RA	131.9	64.43	32.42
	Ours w/o DA	131.9	47.52	29.95
	Ours	131.9	63.58	28.98
BCI IV-2b	Ours-S	94.6	45.17	21.89
	Ours-U	128.2	48.08	13.35
	Ours w/o RA	128.1	58.09	23.05
	Ours w/o DA	128.1	45.75	23.34
	Ours	128.1	60.74	24.72
Zhou2016	TCFormer	125.6	5.89	15.97
	Ours-S [†]	96.8	6.29	13.36
	Ours-U [†]	130.4	11.42	14.65
	Ours w/o RA [†]	130.4	11.66	18.88
	Ours w/o DA	130.3	11.06	31.23
	Ours	130.3	15.20	28.99

6. Ablation Study

Table 7 follows the primary chronological protocol. “w/o DA” removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining RA and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

On BCI IV-2a, removing RA or DA reduced accuracy from 75.81% to 69.06% and 71.37%, respectively; the no-DA difference was significant ($p = 0.0078$). On BCI IV-2b, no-RA (83.61%), no-IM (83.97%; $p = 0.4063$), and no-DA (83.11%; $p = 0.4961$) were effectively comparable with the complete model (83.41%). On Zhou2016, no-RA (84.00%), no-IM (82.33%), and no-DA (76.33%) exceeded the complete model (74.00%); with only four folds, none of these differences was significant ($p = 0.1250, 0.3750, \text{ and } 0.6250$).

Table 7: Ablation results under the primary chronological protocol: mean $\pm$ standard deviation of accuracy (%) and Cohen’s κ ($\times 100$) across target subjects. BCI IV-2a is scored on session 2, BCI IV-2b on sessions 4–5, and Zhou2016 on session 3. RA: Riemannian alignment; DA: Stage 1 domain adaptation; IM: IM-TTA.

Dataset	Metric	Complete	w/o RA	w/o DA	w/o IM
BCI IV-2a	Acc.	75.81 $\pm$ 10.63	69.06 $\pm$ 8.97	71.37 $\pm$ 11.60	72.15 $\pm$ 13.58
	κ	67.70 $\pm$ 14.20	58.70 $\pm$ 12.00	61.80 $\pm$ 15.50	62.90 $\pm$ 18.10
BCI IV-2b	Acc.	83.41 $\pm$ 10.35	83.61 $\pm$ 10.06	83.11 $\pm$ 11.47	83.97 $\pm$ 10.71
	κ	66.80 $\pm$ 20.70	67.20 $\pm$ 20.10	66.20 $\pm$ 22.90	67.90 $\pm$ 21.40
Zhou2016	Acc.	74.00 $\pm$ 6.02	84.00 $\pm$ 3.16	76.33 $\pm$ 3.79	82.33 $\pm$ 3.82
	κ	61.00 $\pm$ 9.03	76.00 $\pm$ 4.70	64.50 $\pm$ 5.70	73.50 $\pm$ 5.70

respectively). Thus, alignment was beneficial on BCI IV-2a but dataset-dependent on BCI IV-2b and Zhou2016.

7. Discussion

The results show that adaptation is dataset dependent. The complete framework clearly surpassed its no-IM variant on BCI IV-2a, whereas the no-IM variant was slightly better on BCI IV-2b. On Zhou2016, the variants without RA or IM-TTA outperformed the complete configuration, while source-only and no-DA training also remained competitive. This outcome may reflect the small four-subject sample, long inter-session gaps, and subject-specific spectral variation. Although filtering to 8–30 Hz and excluding the first 0.5 s focus covariance estimation on MI-related activity rather than broadband drift and cue transients, repeated-seed experiments are still required to establish the stability of the observed RA and IM-TTA sensitivity.

Architecturally, multi-scale and depthwise convolutions preserve local spectral-spatial evidence, while alternating selective scans and the TCN model complementary long- and short-range dynamics. The alternating scans exploit complete offline trials and are non-causal; online use would require forward-only processing and device-specific latency validation.

The method is transductive because IM-TTA accesses unlabeled evaluation-session signals before scoring. Comparisons must therefore provide baselines with identical target access. Other limitations include

the three-channel BCI IV-2b setting, the small Zhou2016 cohort, and the absence of repeated-seed and fully matched baseline results. Participant sex is reported only in aggregate or not at all for the public datasets, and the cohorts are too small for sex- or gender-stratified analysis; potential sex-related differences in decoding performance were therefore not assessed and limit the generalizability of the findings.

8. Conclusion

We presented a cross-subject MI-EEG framework combining multi-scale spatial-temporal convolution, alternating selective state-space blocks, local-global TCN fusion, hierarchical UDA, and constrained IM-TTA. Its chronological LOSO protocol excludes target labels from preprocessing, training, adaptation, and checkpoint selection. Ours obtained $75.81 \pm 10.63\%$ session-2 accuracy on BCI IV-2a and $83.41 \pm 10.35\%$ sessions 4–5 accuracy on BCI IV-2b, with corresponding Cohen’s κ scores of 67.70 ± 14.20 and 66.80 ± 20.70 on the $\times 100$ scale. On Zhou2016, Ours obtained $74.00 \pm 6.02\%$ target-session-3 accuracy and 61.00 ± 9.03 Cohen’s κ . Matched baselines and repeated-seed experiments remain necessary before drawing broader conclusions.

CRedit authorship contribution statement

[Author name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [29] and the Zhou2016 dataset [30] were accessed through the MOABB library [27]; no new data were collected.

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